Lab Programs

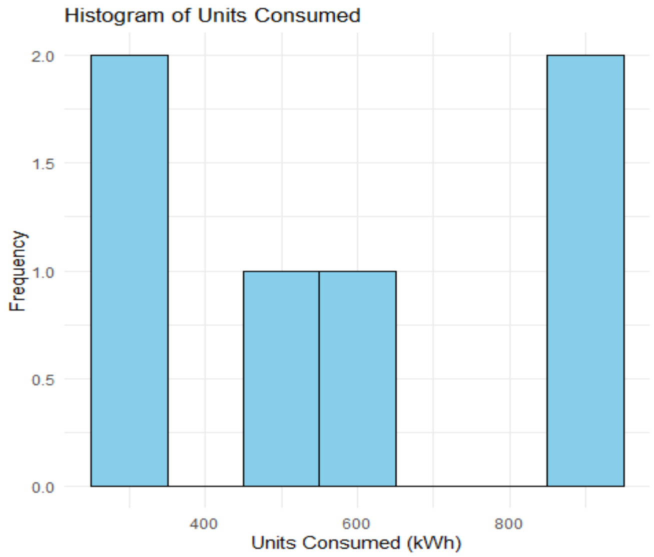
SET 21 - Energy Consumption Data.

21 A. Import Dataset, Histogram & Density Plot

Program: (Histogram)

```
energy_data <- data.frame(
  Sector =
c("Residential","Commercial","Industrial","Residential","Commercial","Industrial"),
  Region = c("North","South","West","East","North","South"),
  Month = c("Jan","Jan","Feb","Feb","Mar","Mar"),
  Temperature = c(15,24,20,18,28,30),
  Units_Consumed = c(320,540,880,350,610,920),
  Cost = c(2100,3600,5900,2300,4100,6200),
  Renewable_Usage = c(22,18,12,25,20,15),
  Peak_Hours = c(4,6,8,5,7,9))
ggplot(energy_data,
aes(x=Units_Consumed))+
geom_histogram(binwidth=100,
fill="skyblue",
color="black")+
labs(title="Histogram of Units Consumed",
x="Units Consumed (kWh)",
y="Frequency")+
theme_minimal()
```

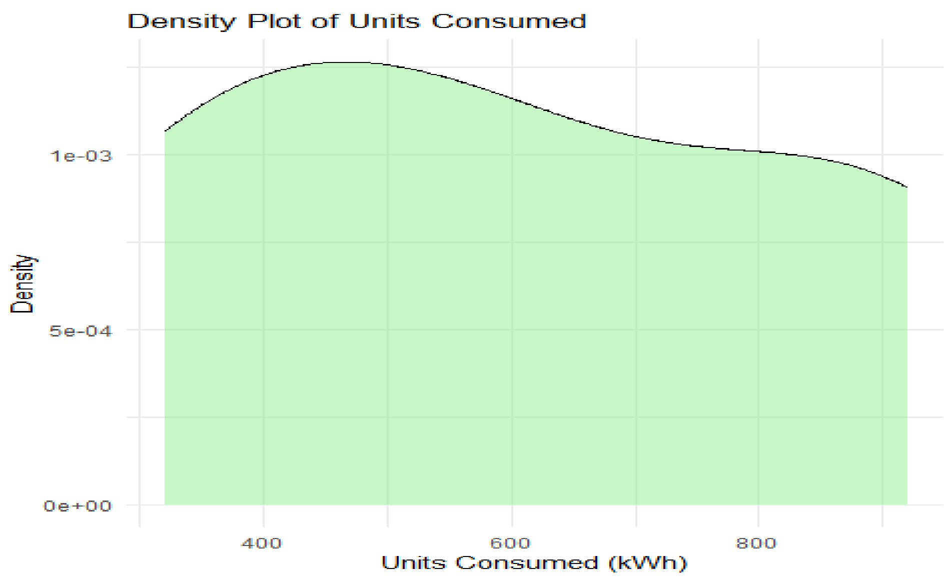
Output:



Program: (Density Plot)

```
ggplot(energy_data,
aes(x=Units_Consumed))+
geom_density(fill="lightgreen",
alpha=0.5)+
labs(title="Density Plot of Units Consumed",
x="Units Consumed (kWh)",
y="Density")+
theme_minimal()
```

Output:

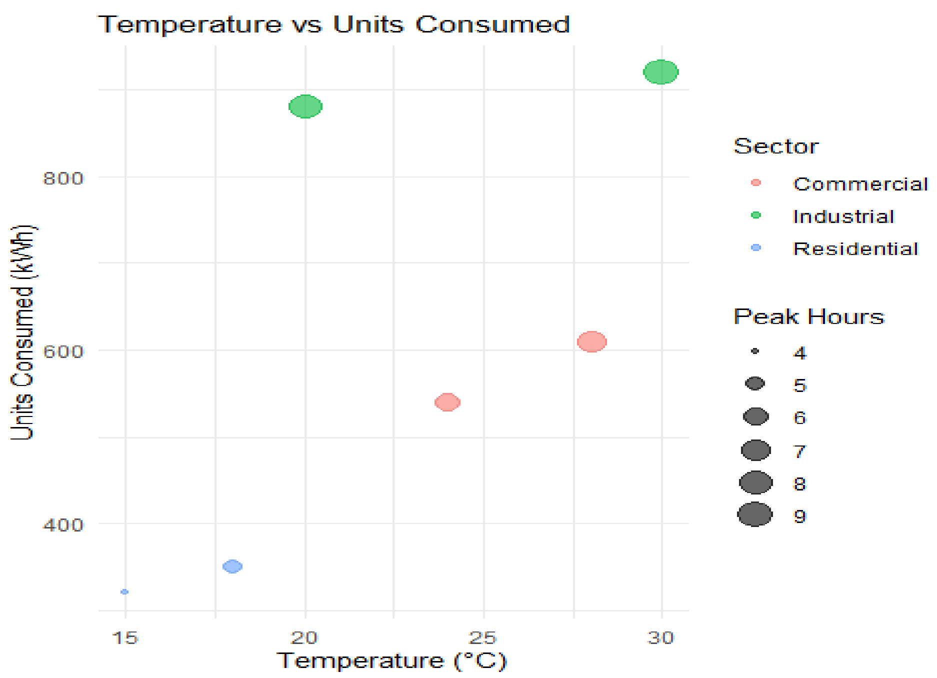


21 B. Scatter Plot

Program:

```
ggplot(energy_data,
aes(x=Temperature,
y=Units_Consumed,
size=Peak_Hours,
color=Sector))+
geom_point(alpha=0.6)+
labs(title="Temperature vs Units Consumed",
x="Temperature (°C)",
y="Units Consumed (kWh)",
size="Peak Hours")+
theme_minimal()
```

Output:



21 C. Average Renewable Usage by Sector and Bar Chart

Program:

```
library(dplyr)
sector_avg <- energy_data %>%
group_by(Sector) %>%
```

summarise(

Average_Renewable = mean(Renewable_Usage)

)sector_avg

Output:

```
# A tibble: 3 × 2
  Sector      Average_Renewable
  <chr>                   <dbl>
1 Commercial               19
2 Industrial               13.5
3 Residential              23.5
```

Program:

ggplot(sector_avg,

aes(x=Sector,

y=Average_Renewable,

fill=Sector))+

geom_bar(stat="identity")+

labs(title="Average Renewable Usage by Sector",

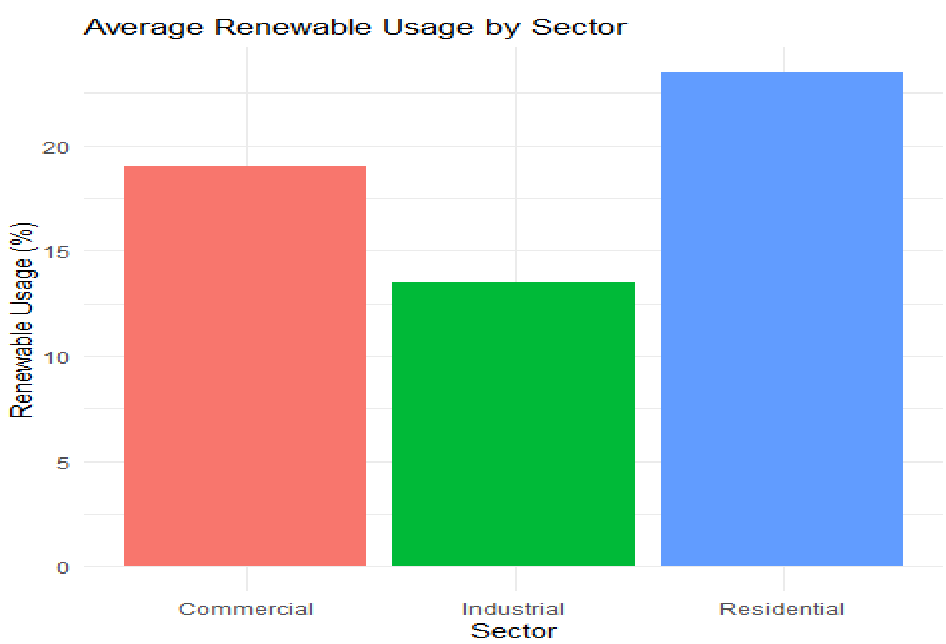
x="Sector",

y="Renewable Usage (%)")+

theme_minimal()+

theme(legend.position="none")

Output:



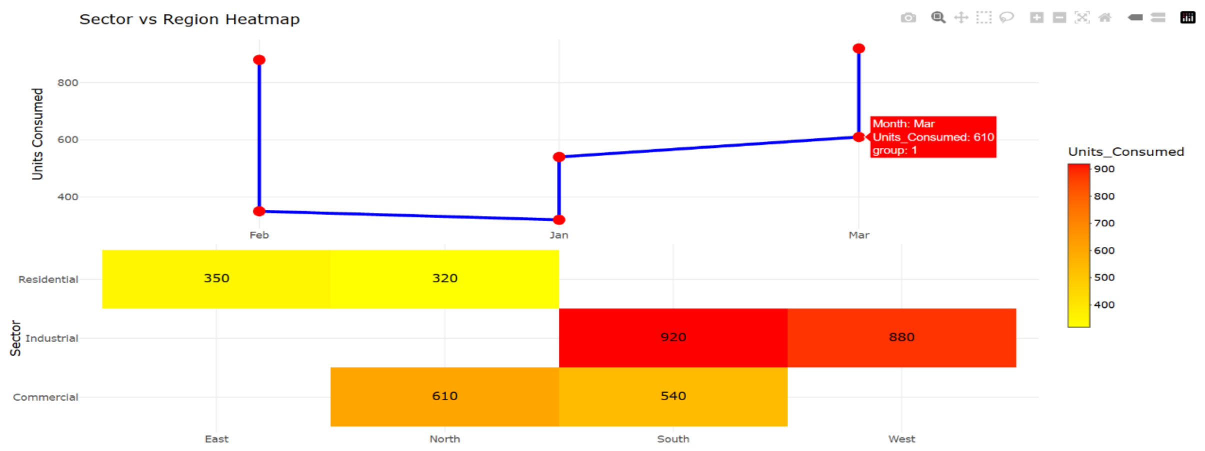
21 D. Tableau Dashboard

Program:

```
library(ggplot2)
library(plotly)
line_plot <- ggplotly(
  ggplot(energy_data,
         aes(x=Month,
             y=Units_Consumed,
             group=1))+
    geom_line(color="blue",linewidth=1)+
    geom_point(size=3,color="red")+
    labs(title="Monthly Energy Usage Trend",
         x="Month",
         y="Units Consumed")+
    theme_minimal())
heat_plot <- ggplotly(
  ggplot(energy_data,
         aes(x=Region, y=Sector,
             fill=Units_Consumed))+
    geom_tile(color="white")+
    geom_text(aes(label=Units_Consumed),color="black")+
    scale_fill_gradient(low="yellow",high="red")+
    labs(title="Sector vs Region Heatmap",
         x="Region",
         y="Sector")+
    theme_minimal())
dashboard <- subplot(
  line_plot,
  heat_plot,
  nrows=2,
```

```
  shareX=FALSE,
  shareY=FALSE,
  titleY=TRUE)
dashboard
```

Output:



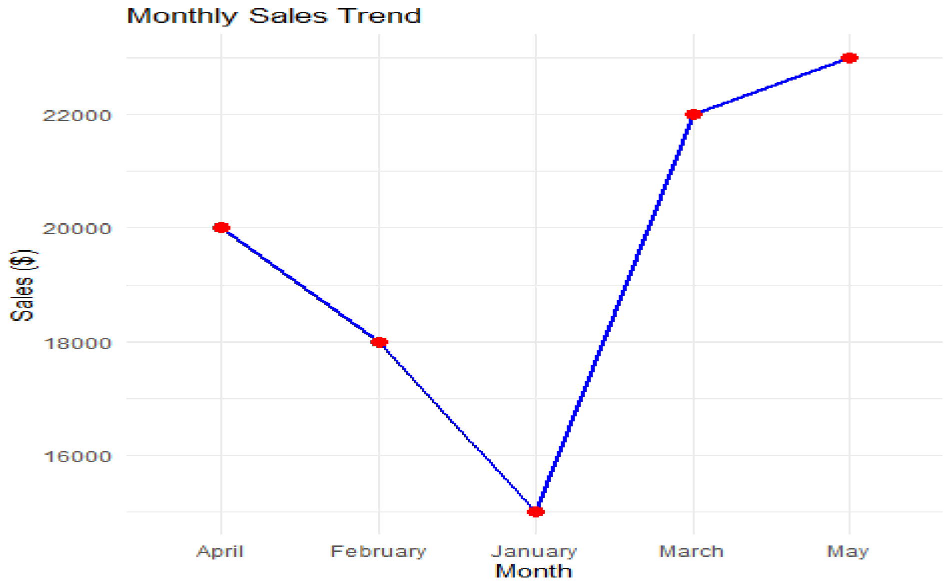
SET 22 - Time Series Analysis.

22 A. Time Series Line Chart

Program:

```
sales_data <- data.frame(
Month=c("January","February","March","April","May"),
Sales=c(15000,18000,22000,20000,23000))
ggplot(sales_data,
aes(x=Month,
y=Sales,
group=1))+
geom_line(color="blue",linewidth=1)+
geom_point(color="red",size=3)+
labs(title="Monthly Sales Trend",
x="Month",
y="Sales ($)")+
theme_minimal()
```

Output:

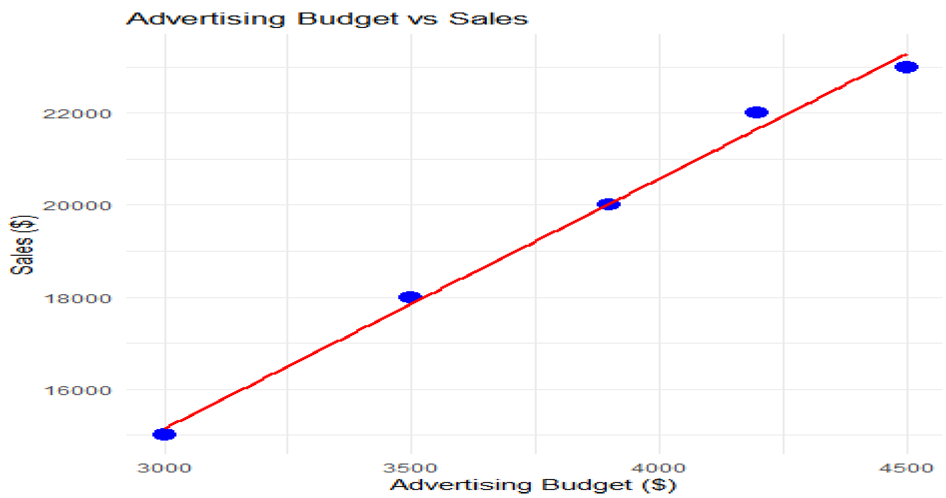


22 B. Scatter Plot

Program:

```
sales_data$Advertising_Budget <- c(3000,3500,4200,3900,4500)
ggplot(sales_data,
aes(x=Advertising_Budget,y=Sales))+
geom_point(size=4,color="blue")+
geom_smooth(method="lm",se=FALSE,color="red")+
labs(title="Advertising Budget vs Sales",
x="Advertising Budget ($)",
y="Sales ($)")+
theme_minimal()
```

Output:

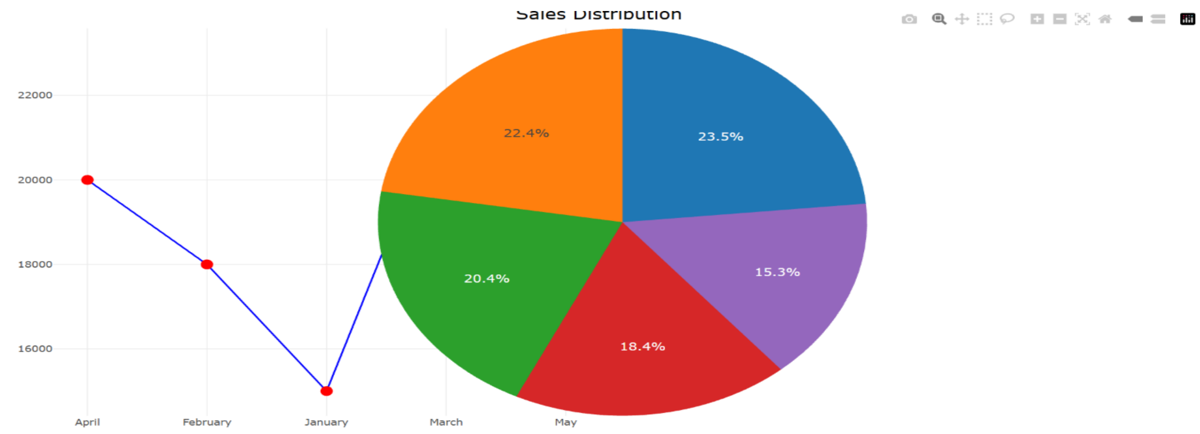


22 C. Tableau Dashboard

Program:

```
line_plot <- ggplotly(
ggplot(sales_data,
aes(x=Month,
y=Sales,
group=1))+
geom_line(color="blue")+
geom_point(color="red",size=3)+
labs(title="Sales Trend")+
theme_minimal())
pie_plot <- plot_ly(
sales_data,
labels=~Month,
values=~Sales,
type="pie") %>%
layout(title="Sales Distribution")
dashboard <- subplot(
line_plot,pie_plot,nrows=1,
shareX=FALSE,
shareY=FALSE)
dashboard
```

Output:

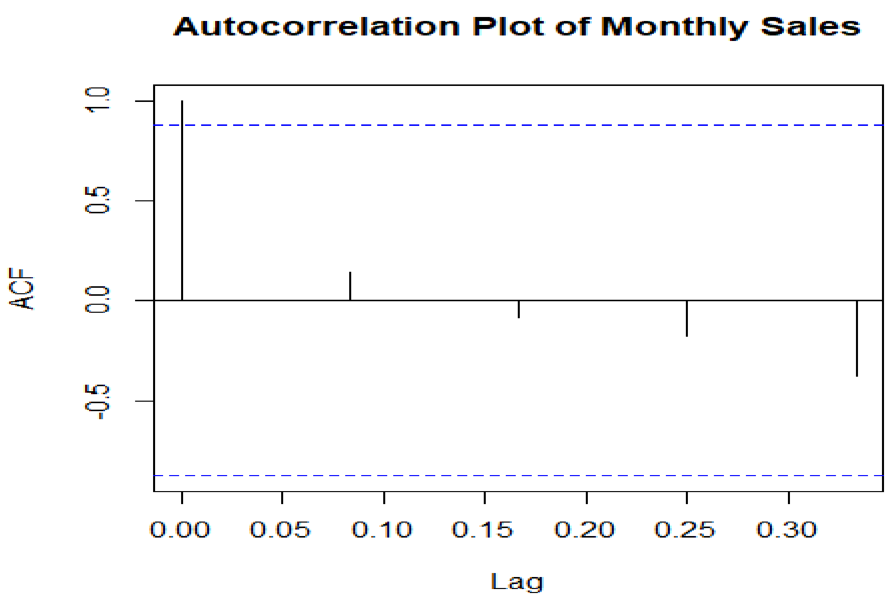


22 D. Autocorrelation Plot

Program:

```
 sales_ts <- ts(
sales_data$Sales,
frequency=12)
acf(
sales_ts,
main="Autocorrelation Plot of Monthly Sales")
```

Output:



SET 23 - Employee Performance Analysis

23 A. Bar Chart: Distribution of Employees Across Departments

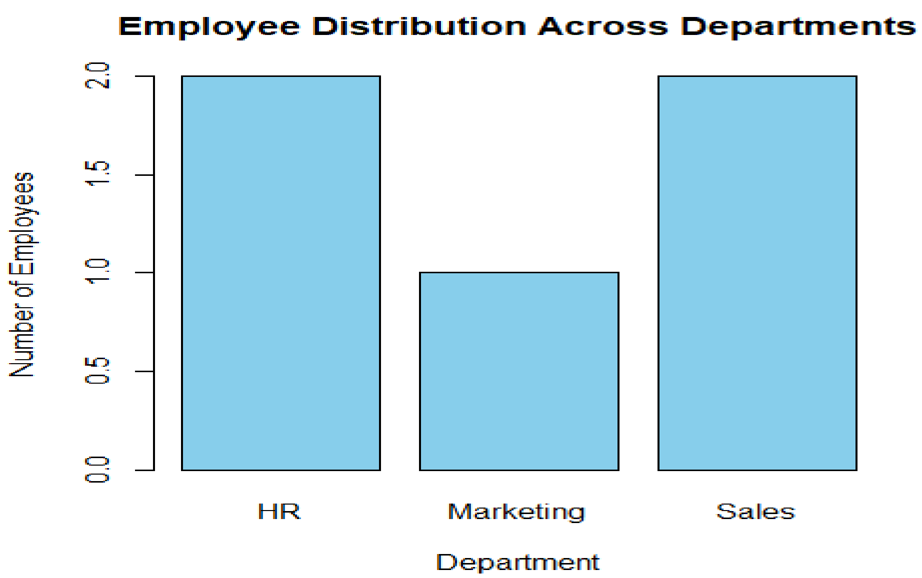
Program:

```
# Create Employee Performance Dataset
employee_data <- data.frame(
  Employee_ID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  Years_of_Service = c(5,3,7,4,2),
  Performance_Score = c(85,92,78,90,76))

# Display dataset
```

```
print(employee_data)
dept_count <- table(employee_data$Department)
barplot(
  dept_count,
  main = "Employee Distribution Across Departments",
  xlab = "Department",
  ylab = "Number of Employees",
  col = "skyblue",
  border = "black")
# Add values on bars
text(
  x = seq_along(dept_count),
  y = dept_count,
  labels = dept_count,
  pos = 3)
```

Output:



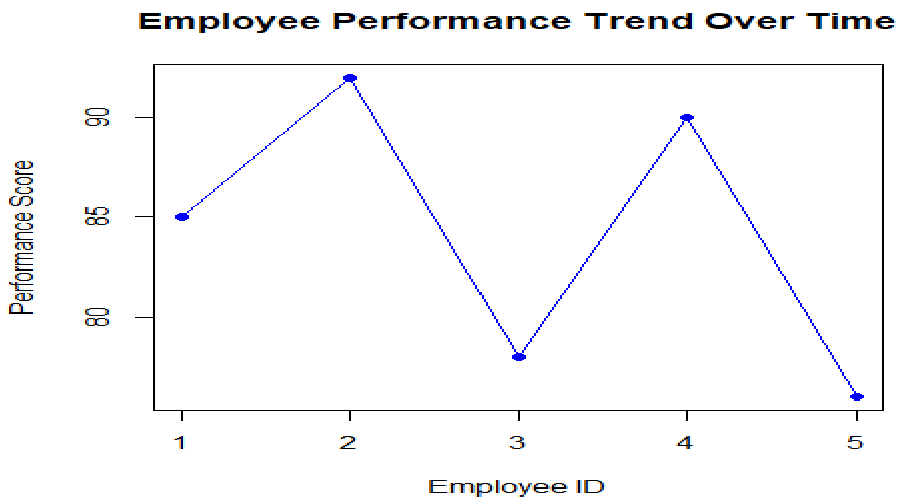
23 B. Line Chart: Employee Performance Trend Over Time

Program:

```
 plot(
 employee_data$Employee_ID,
```

```
  employee_data$Performance_Score,
  type = "o",
  main = "Employee Performance Trend Over Time",
  xlab = "Employee ID",
  ylab = "Performance Score",
  col = "blue",
  pch = 16)
  grid()
```

Output:



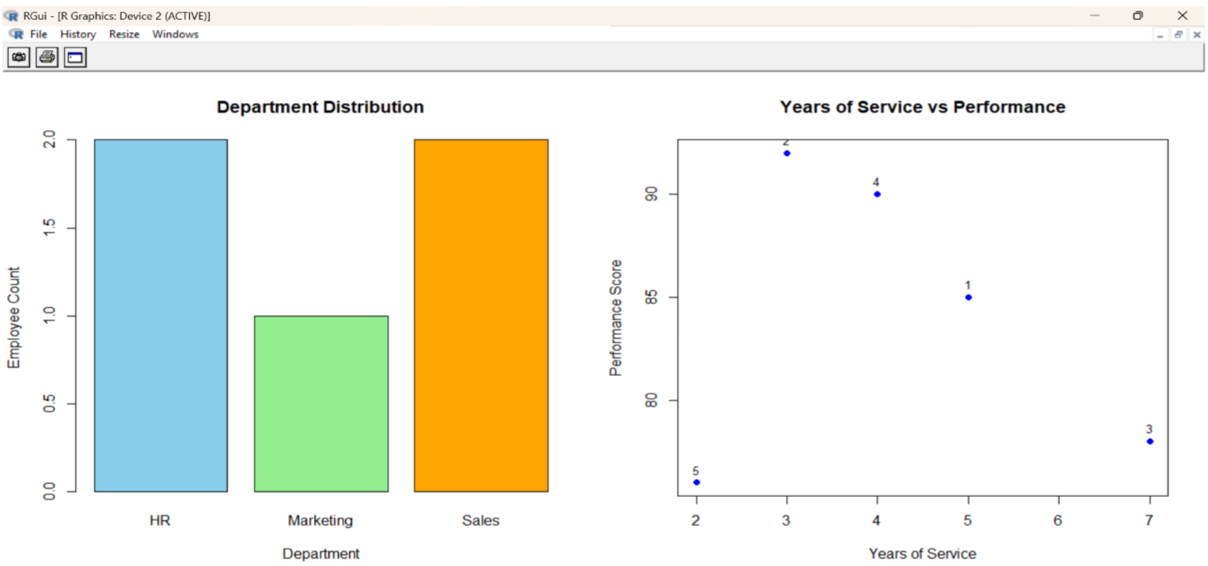
23 C. Tableau Dashboard: Employee Performance Analysis

Program:

```
# Employee Performance Dataset
employee_data <- data.frame(
  Employee_ID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  Years_of_Service = c(5,3,7,4,2),
  Performance_Score = c(85,92,78,90,76))
# Divide plotting area into 1 row and 2 columns
par(mfrow = c(1,2))
dept_count <- table(employee_data$Department)
barplot(
```

```
  dept_count,
  main = "Department Distribution",
  xlab = "Department",
  ylab = "Employee Count",
  col = c("skyblue","lightgreen","orange"))
  plot(
  employee_data$Years_of_Service,
  employee_data$Performance_Score,
  main = "Years of Service vs Performance",
  xlab = "Years of Service",
  ylab = "Performance Score",
  pch = 19,
  col = "blue")
  text(
  employee_data$Years_of_Service,
  employee_data$Performance_Score,
  labels = employee_data$Employee_ID,
  pos = 3,
  cex = 0.8)
par(mfrow = c(1,1))
```

Output:



23 D. Performance Data Table

Program:

```
# Display performance table
performance_table <- data.frame(
  Employee_ID = employee_data$Employee_ID,
  Department = employee_data$Department,
  Years_of_Service = employee_data$Years_of_Service,
  Performance_Score = employee_data$Performance_Score)
# Print table
print(performance_table)
```

Output:

```
  Employee_ID Department Years_of_Service Performance_Score
1           1      Sales                5                85
2           2         HR                3                92
3           3  Marketing                7                78
4           4      Sales                4                90
5           5         HR                2                76
>
```

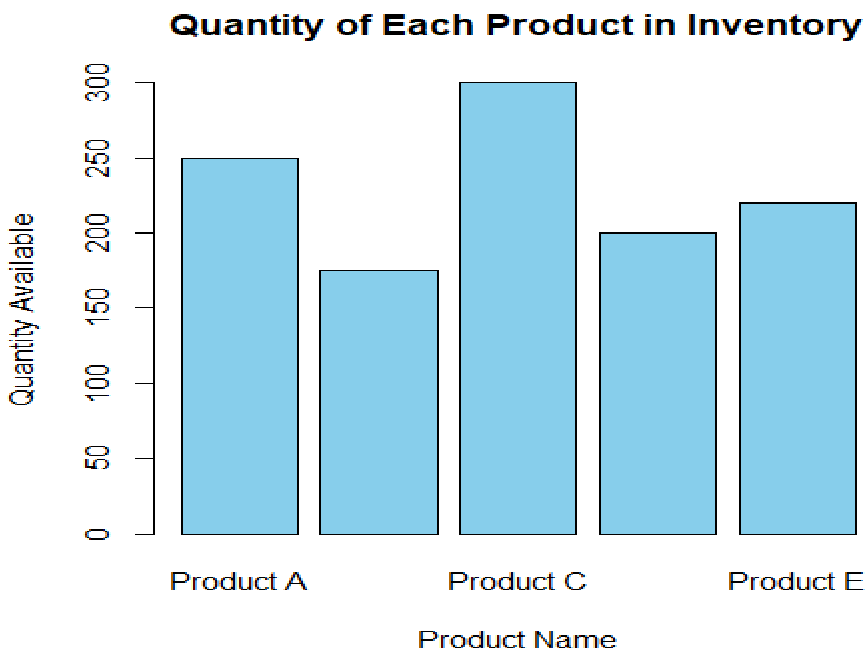
SET 24 - Product Inventory Management

24 A.

Program:

```
inventory <- data.frame(
Product_ID = c(1,2,3,4,5),
Product_Name = c("Product A","Product B","Product C","Product D","Product E"),
Quantity = c(250,175,300,200,220))
print(inventory)
barplot(
  inventory$Quantity,
  names.arg = inventory$Product_Name,
  col = "skyblue",
  main = "Quantity of Each Product in Inventory",
  xlab = "Product Name",
  ylab = "Quantity Available")
```

Output:

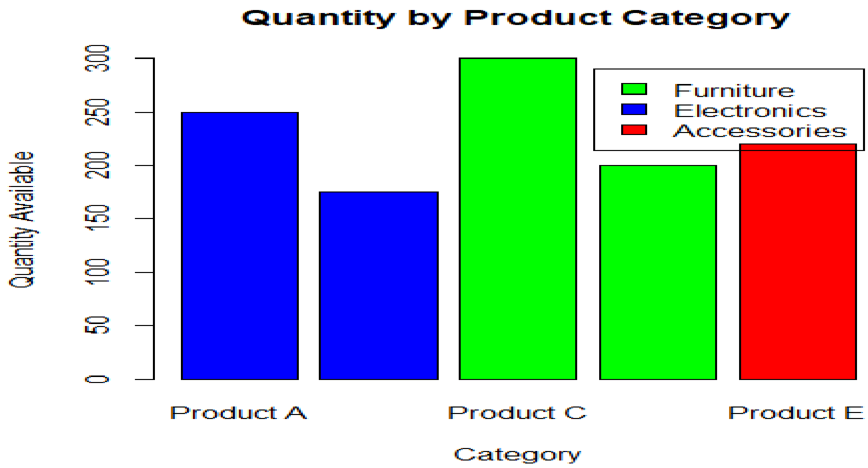


24 B. Stacked Bar Chart – Quantity by Product Category

Program:

```
# Add categories
inventory$Category <- c("Electronics","Electronics",
                        "Furniture","Furniture",
                        "Accessories")
# Create table
stack_data <- xtabs(Quantity ~ Category + Product_Name, data = inventory)
# Stacked bar chart
barplot(
  stack_data,
  col = c("red","blue","green","orange","purple"),
  main = "Quantity by Product Category",
  xlab = "Category",
  ylab = "Quantity Available",
  legend = rownames(stack_data))
```

Output:



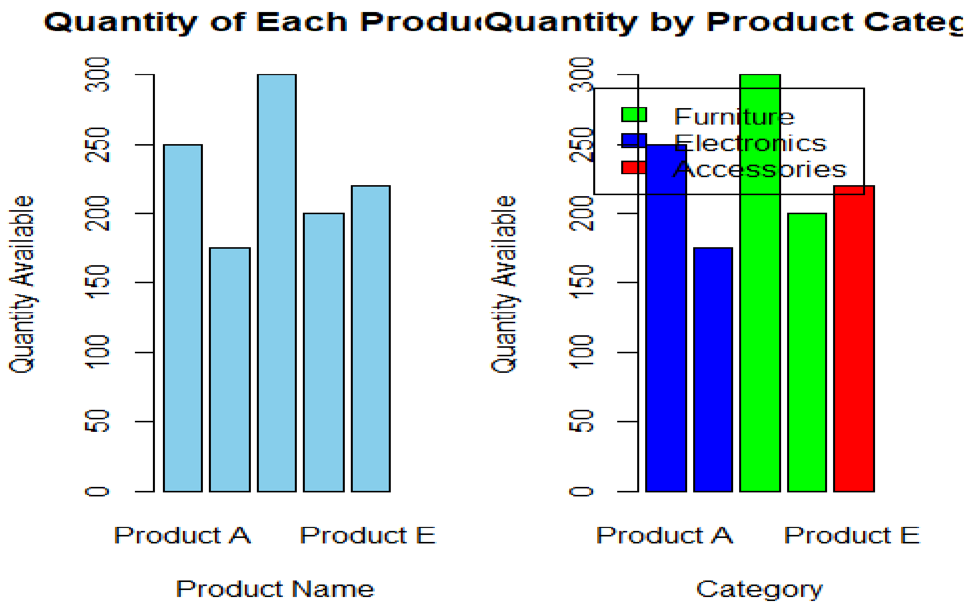
24 C. Tableau Dashboard

Program:

```
# Product Inventory Dataset
 inventory <- data.frame(
 Product_ID = c(1,2,3,4,5),
 Product_Name = c("Product A","Product B","Product C","Product D","Product E"),
 Quantity = c(250,175,300,200,220),
 Category = c("Electronics","Electronics","Furniture","Furniture","Accessories"))
  # Arrange plots side by side (Dashboard)
  par(mfrow = c(1,2))
  # -------------------------
# Bar Chart
# -------------------------
 barplot(
 inventory$Quantity,
 names.arg = inventory$Product_Name,
 col = "skyblue",
 main = "Quantity of Each Product",
 xlab = "Product Name",
 ylab = "Quantity Available")
 stack_data <- xtabs(Quantity ~ Category + Product_Name, data = inventory)
```

```
barplot(
stack_data,
col = c("red","blue","green","orange","purple"),
main = "Quantity by Product Category",
xlab = "Category",
ylab = "Quantity Available",
legend.text = rownames(stack_data))
par(mfrow = c(1,1))
```

Output:



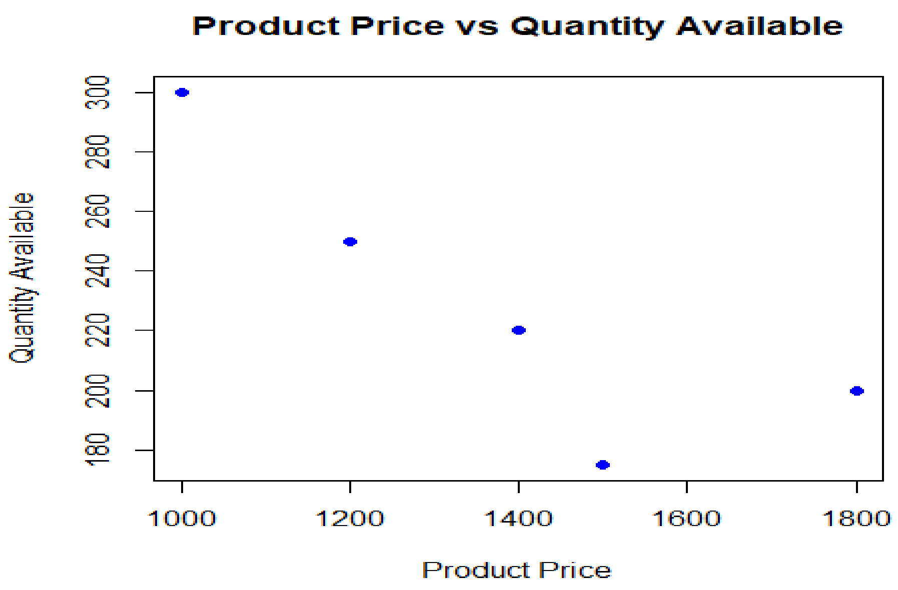
24 D. Scatter Plot – Product Price vs Quantity Available

Program:

```
inventory$Price <- c(1200,1500,1000,1800,1400)
 plot(
inventory$Price,
inventory$Quantity,
pch = 19,
col = "blue",
main = "Product Price vs Quantity Available",
xlab = "Product Price",
```

```
  ylab = "Quantity Available")
  text(
  inventory$Price,
  inventory$Quantity,
  labels = inventory$Product_Name,
  pos = 3,
  cex = 0.8)
```

Output:



SET 25 - Website Traffic Analysis

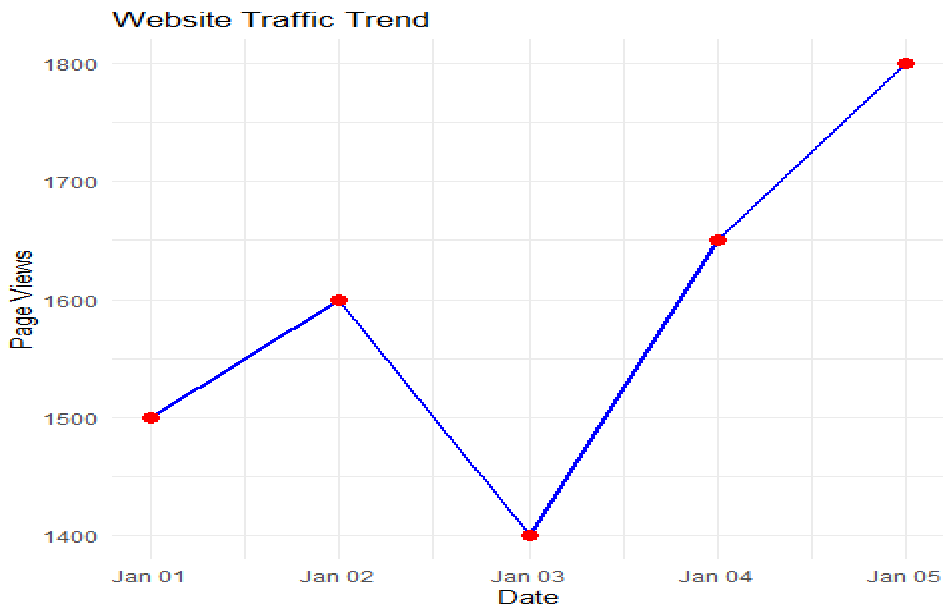
25 A. Line Chart

Program:

```
traffic_data <- data.frame(
  Date = as.Date(c("2023-01-01",
                   "2023-01-02",
                   "2023-01-03",
                   "2023-01-04",
                   "2023-01-05")),
  PageViews = c(1500,1600,1400,1650,1800),
  CTR = c(2.3,2.7,2.0,2.4,2.6))
```

```
ggplot(traffic_data,
       aes(x=Date,
           y=PageViews,
           group=1))+
  geom_line(color="blue",linewidth=1)+
  geom_point(color="red",size=3)+
  labs(title="Website Traffic Trend",
       x="Date",
       y="Page Views")+
  theme_minimal()
```

Output:



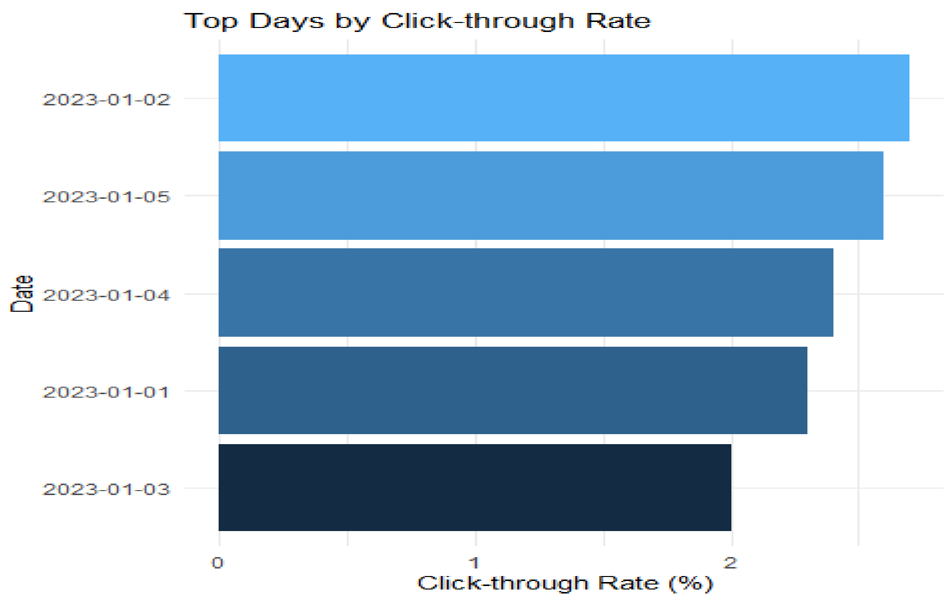
25 B. Bar Chart

Program:

```
top_ctr <- traffic_data[order(-traffic_data$CTR),]
ggplot(top_ctr,
       aes(x=reorder(as.character(Date),CTR),
           y=CTR,
           fill=CTR))+
  geom_bar(stat="identity")+
  coord_flip()+
```

```
  labs(title="Top Days by Click-through Rate",
       x="Date",
       y="Click-through Rate (%)")+
  theme_minimal()+
  theme(legend.position="none")
```

Output:



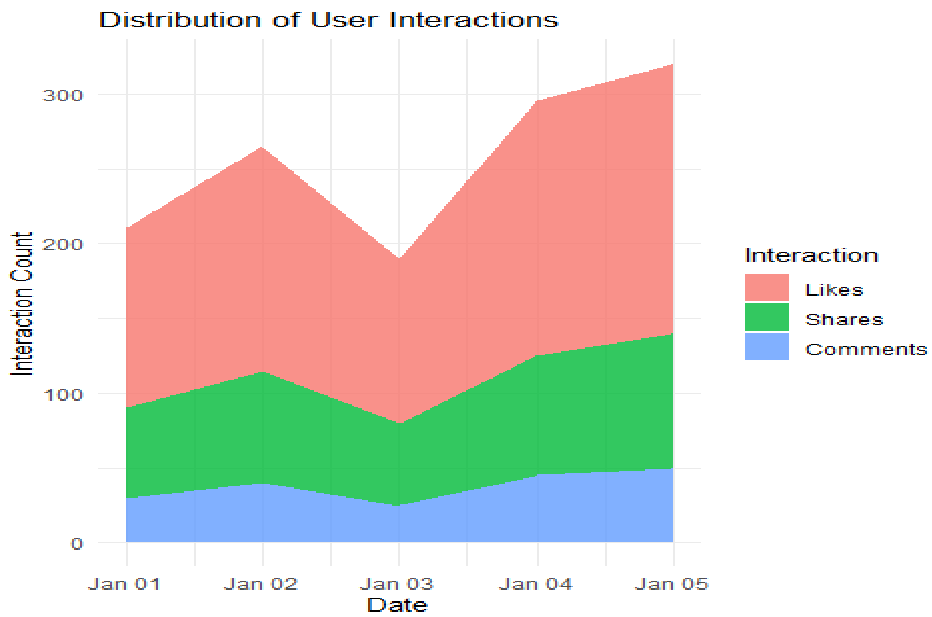
25 C. Stacked Area Chart

Program:

```
interaction_data <- data.frame(
  Date=as.Date(c("2023-01-01",
                 "2023-01-02",
                 "2023-01-03",
                 "2023-01-04",
                 "2023-01-05")),
  Likes=c(120,150,110,170,180),
  Shares=c(60,75,55,80,90),
  Comments=c(30,40,25,45,50))
  id.vars="Date",
  variable.name="Interaction",
  value.name="Count")
```

```
ggplot(interaction_long,
      aes(x=Date,y=Count,
          fill=Interaction,
          group=Interaction))+
  geom_area(alpha=0.8)+
  labs(title="Distribution of User Interactions",
      x="Date", y="Interaction Count")+
  theme_minimal()
```

Output:



25 D. Interactive Dashboard (R Equivalent of Tableau)

Program:

```
line_plot <- ggplotly(
  ggplot(traffic_data,
        aes(x=Date,
            y=PageViews,
            group=1))+
    geom_line(color="blue")+
    geom_point(color="red",size=3)+
    labs(title="Website Traffic Trend")+
    theme_minimal())
```

```
source_data <- data.frame(
  Source=c("Direct","Search","Social","Referral"),
  PageViews=c(800,450,350,250)
bar_plot <- ggplotly(
  ggplot(source_data,
         aes(x=Source,
             y=PageViews,
             fill=Source))+
    geom_bar(stat="identity")+
    labs(title="Page Views by Source")+
    theme_minimal()+
    theme(legend.position="none"))
dashboard <- subplot(
  line_plot,
  bar_plot,
  nrows=1,
  shareX=FALSE,shareY=FALSE)
dashboard
```

Output: